Chapter 5. Instruction Set Architecture

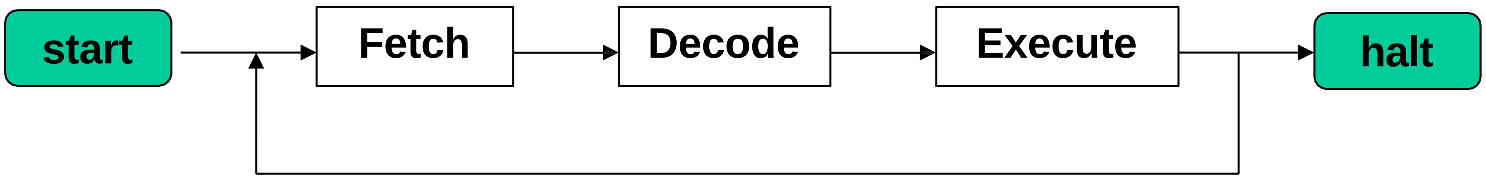
Instruction Cycle

Timing Mechanism
Fetch and Decode Phases of the Instruction Cycle
Execute Phase of the Command Cycle

Microprocessing Steps for Immediate, Direct and Indirect Mode of ADD Command

Instruction Cycle

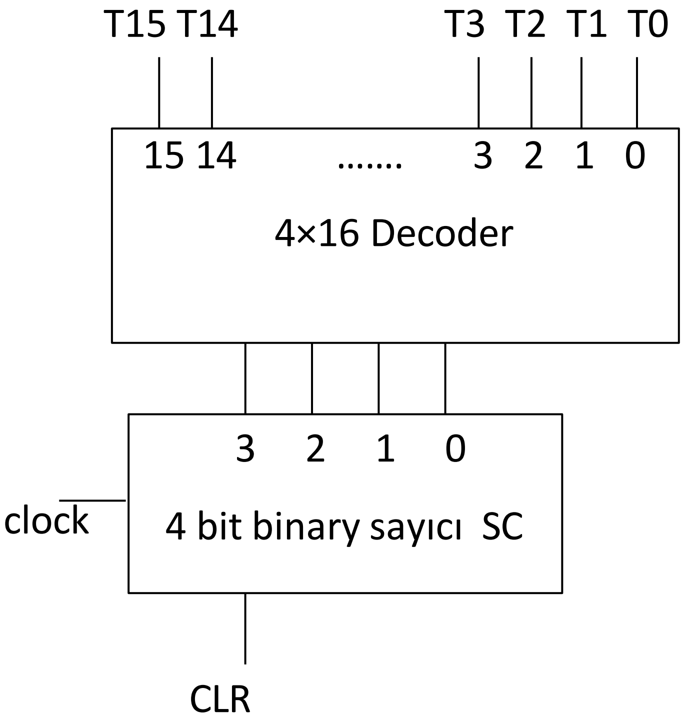
The computer is expected to process the commands in memory sequentially. The set of operations that a command must perform is called the command cycle. These consist of 3 stages with a command cycle; fetch, decode and execute phases. While the opcode part of the instruction is taken from memory and put into the instruction register (IR), the part related to the addressing mode is applied as input to the addressing mode decoder, and the part related to the operation type is applied as input to the instruction decoder. Then, the signals received from these decoders activate the mechanism that initiates the necessary sub-processes (microprocesses) for the relevant command.



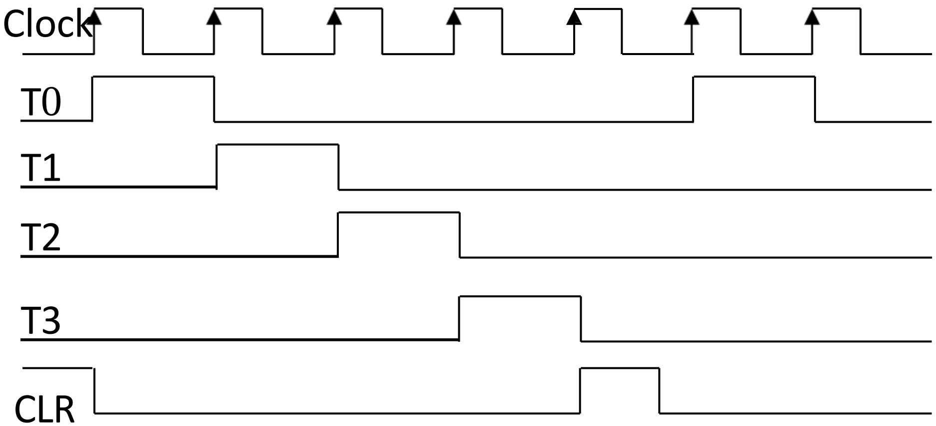
- Instruction/Command Cycle -

Timing Mechanism

Since a computer system is expected to sequentially operate the commands in memory and the microprocesses of each command, a hardware mechanism is needed.



- Timing mechanism -



- Timing Signals -

Fetch and Decode Phases of the Instruction Cycle

Sub-processes of the command consist of a different number of steps depending on the type of command. But the fetch and decode phases of all commands have the same structure and consist of 3 steps. The timing of these steps is accomplished via a counter. **The fetch phase** is performed in two steps (T0 and T1). **The decode phase** is a single step and is performed in the T2 time period. **The Execute phase** starts from the T3 time period and consists of a various number of steps depending on the type of command. However, since instructions with natural addressing mode are 1-byte instructions, operations in the T2 time period will not be performed.

T0 : AR ← PC

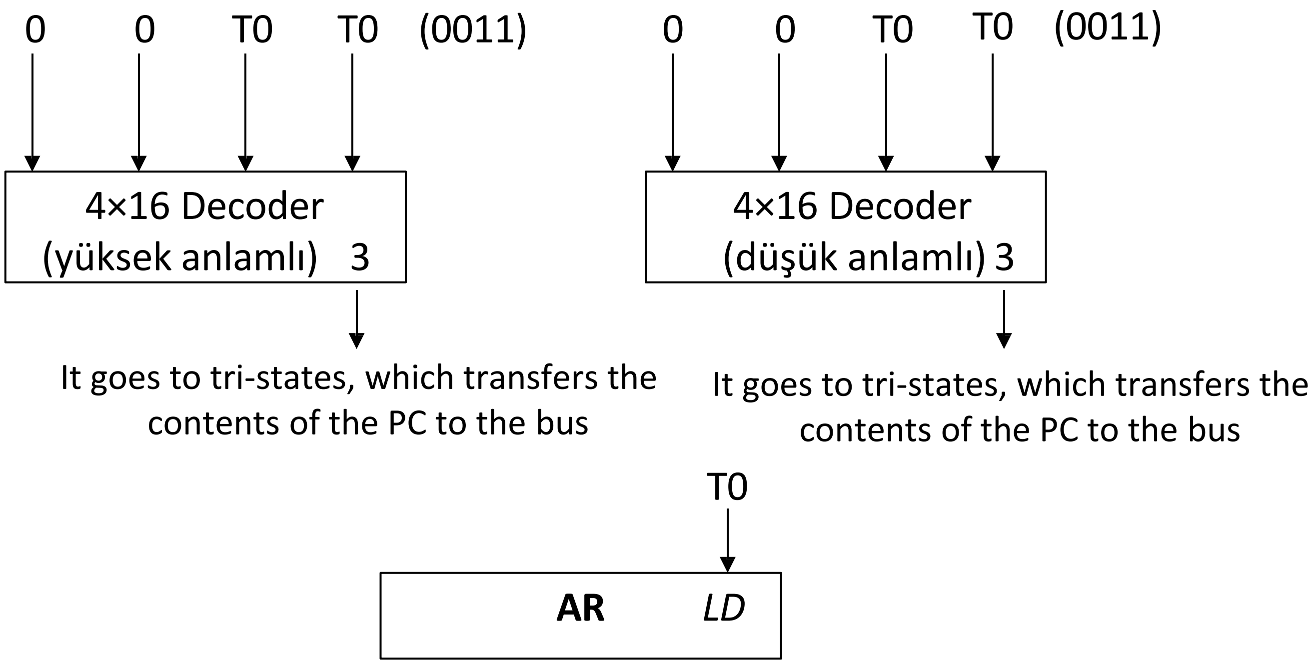
T1 : IR ← M[AR], PC ← PC+1

T2. $\overline{ADRMD0}$: AR←PC, PC ← PC+1, decoding phase

T0 Time Slot

In time slot T0, the address of the next command is loaded into the address register via the PC. For this process, the LD input of the address register is activated and the content of the PC is transferred to the system bus.

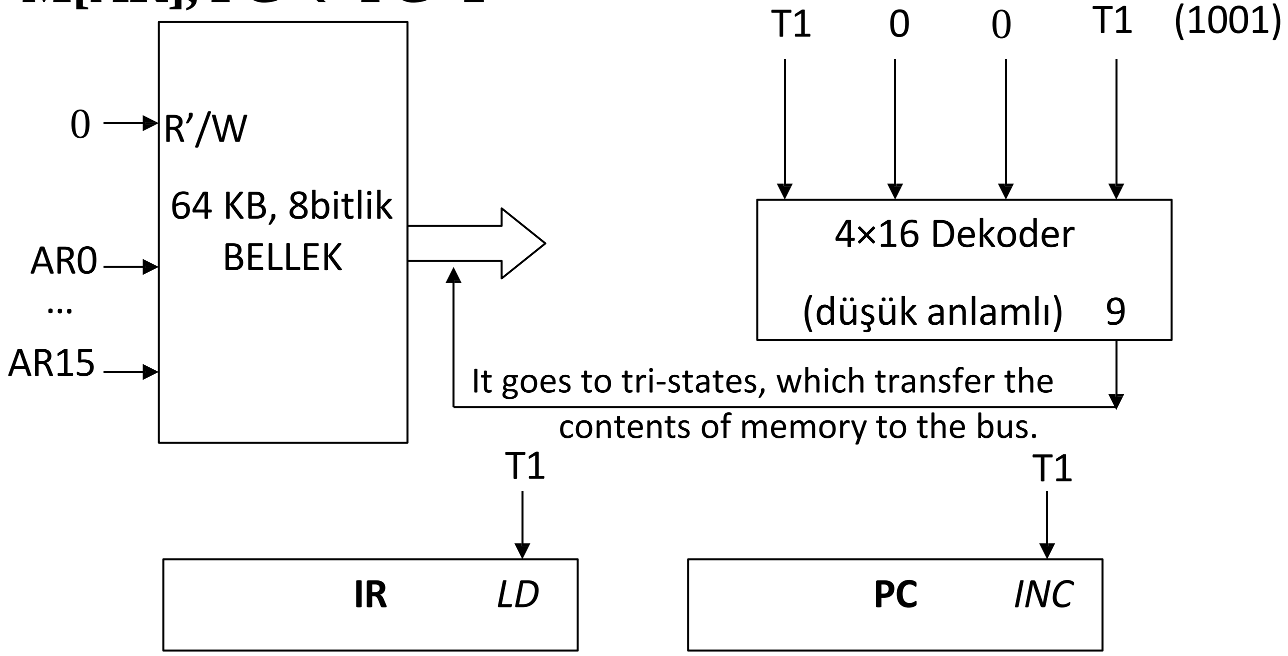
T0: AR ← PC



T1 Time Slot

In time period T1, the opcode of the instruction in the memory region specified by the address register is transferred to the system bus. By activating the LD input of the instruction register, the opcode is transferred to the instruction register. The contents of the program counter are incremented by 1 (to indicate the next instruction if the native addressing mode is used, or the address information of the operand in other addressing modes).

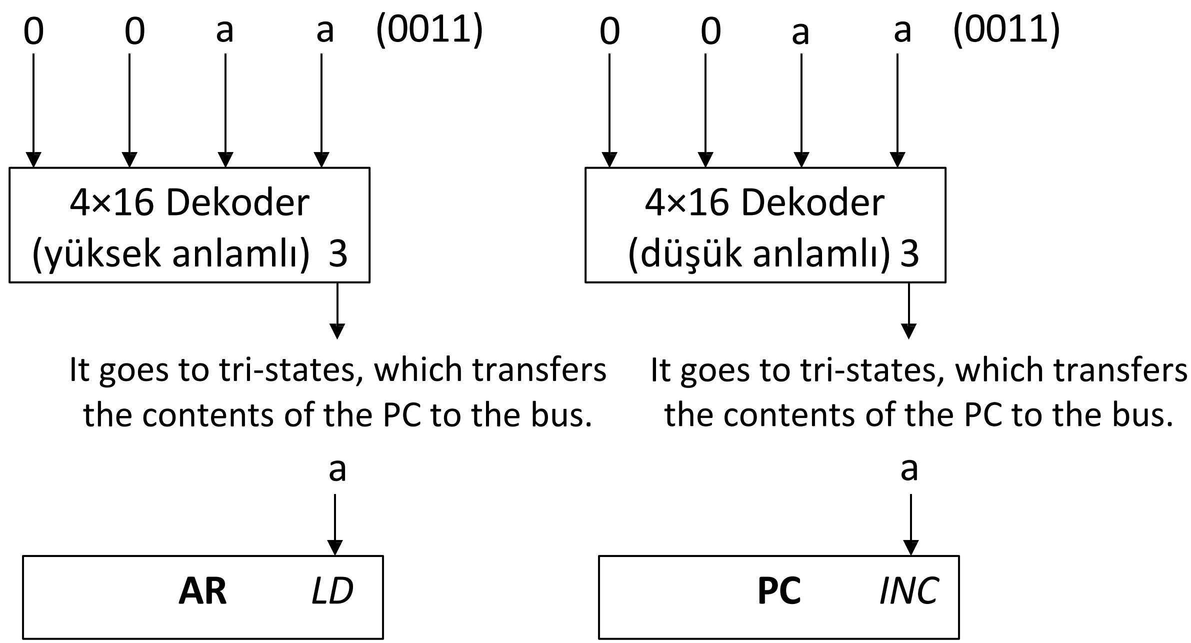
T1: IR ← M[AR], PC ← PC+1



T2 Time Slot

In time period T2, the contents of the PC are transferred to the AR. If it is PC, it is again increased by 1; If it is a 2 byte command, to show the next command, or if it is a 3 byte command, to show the low-meaning address information in the address part of the command.

$\underbrace{\text{T2. } \overline{ADRMD0}}_{\mathbf{a}}$: AR←PC, PC ← PC+1, decode phase

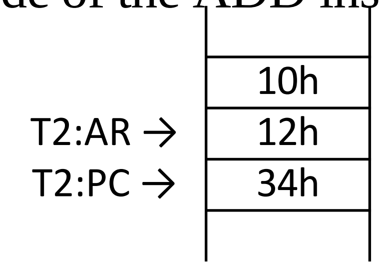


Execute Phase of the Command Cycle

As mentioned before, how many steps the execution phase takes varies depending on the type of command and addressing mode.

Example: Microoperation steps of the ADD instruction with immediate addressing mode.

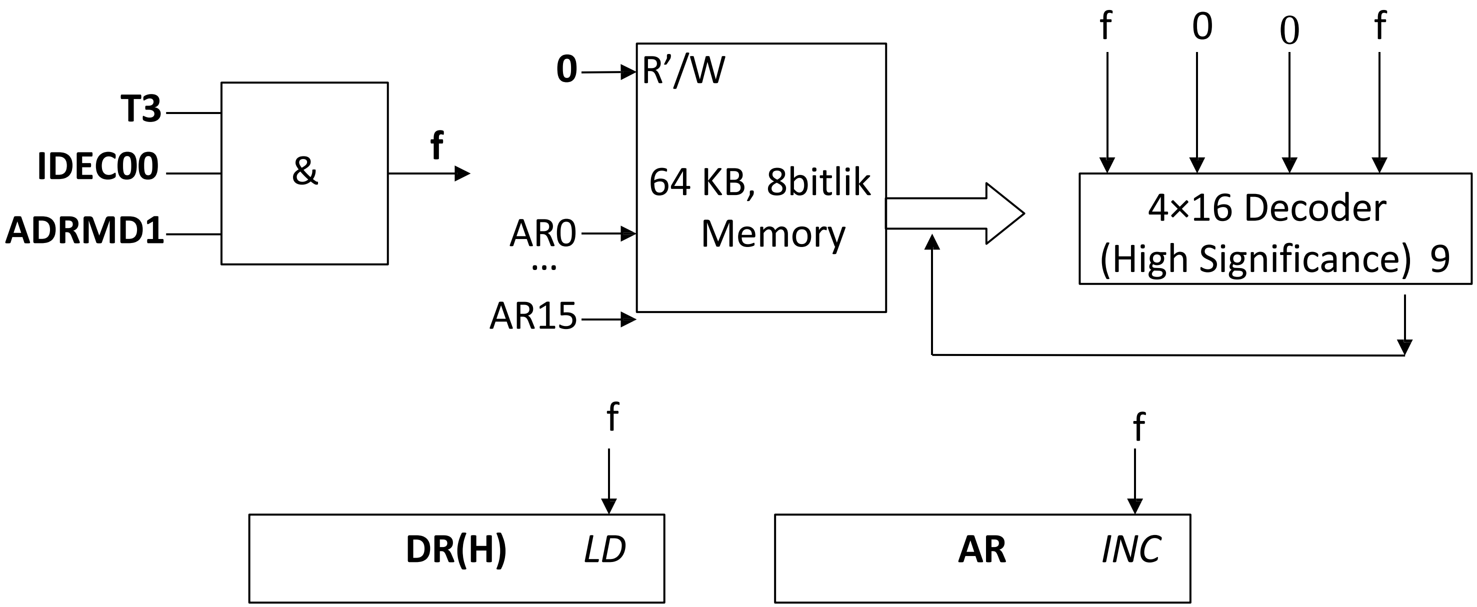
Like the ADD #1234h command. Here the # sign indicates immediate mode. It is a 3 byte command. The opcode of the ADD instruction is $0\ 001\ 0000_2$ = 10h

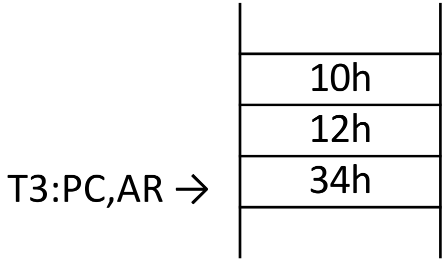


In immediate addressing mode, the ACC was activated with a fixed number (1234h). The number in the ACC and memory will be added and the result will be transferred to the ACC. Currently the AR logger shows the high significance part of our number in memory. We will need to put this value in the high significant part of our DR register. Then, we will need to increase the AR register by 1 to access the low-significant part of our number.

T3 Step For Immediate Mode

T3. IDEC00.ADRMD1: $DR_H \leftarrow M[AR]$, $AR \leftarrow AR+1$

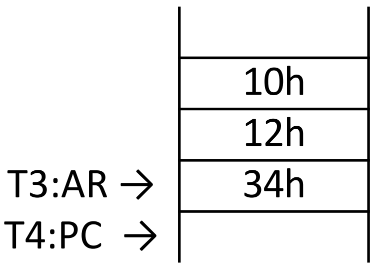




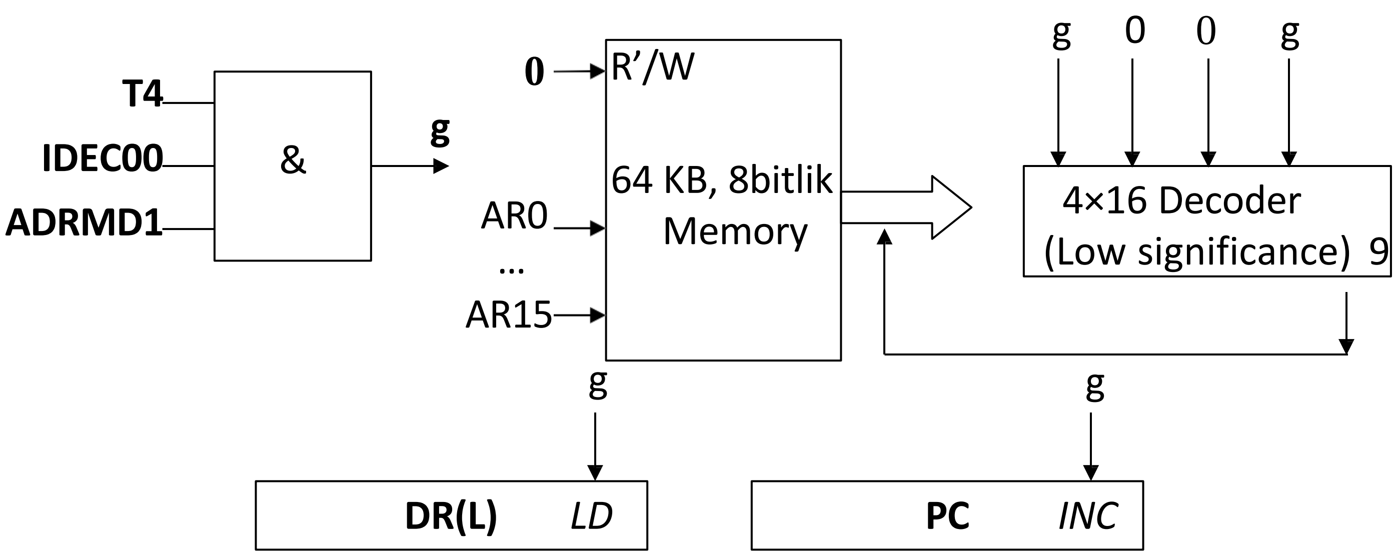
It could have been considered to use AR ← PC instead of AR ← AR+1 operation in step T3, however, two devices cannot use the bus at the same time.

T4 Step For Immediate Mode

T4. IDEC00.ADRMD1: $DR_L \leftarrow M[AR]$, $PC \leftarrow PC+1$



In the previous steps of the ADD command with immediate mode, PC was incremented 2 times. Since this command is 3 bytes long, it is increased once more so that the PC displays the next command.

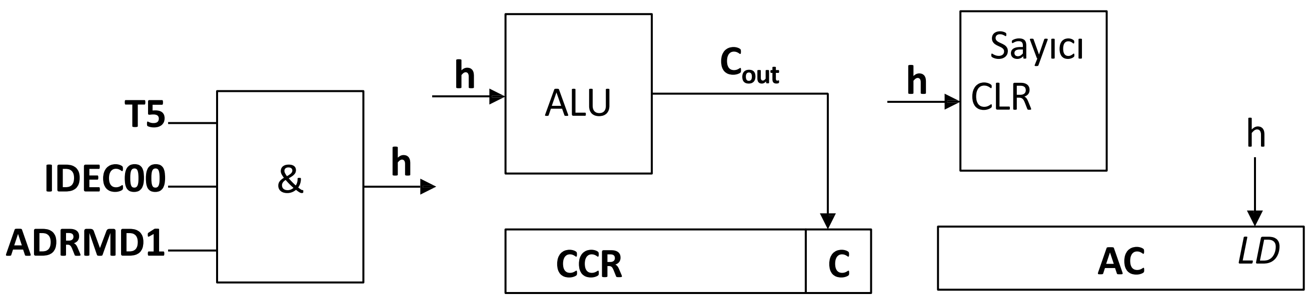


T5 Step For Immediate Mode

In step T5, the necessary action will be performed on the ALU and the counter will be reset:

T5. IDEC00.ADRMD1: AC $\leftarrow$ AC+DR, C $\leftarrow$ C_{out}, SC $\leftarrow$ 0

In the T5 timing period, the necessary addition operation was performed in the ALU, and the result of the operation was transferred to the AC by activating the LD input of the AC. The carry flag has been updated since a hand may occur as a result of the addition process. Additionally, the counter is reset for the fetch operation of the next instruction.



Example: Microoperation steps of ADD instruction with direct addressing mode

Like the ADD 1000h command. There is no sign in front of the address section here, it indicates direct addressing mode. It is a 3 byte command.

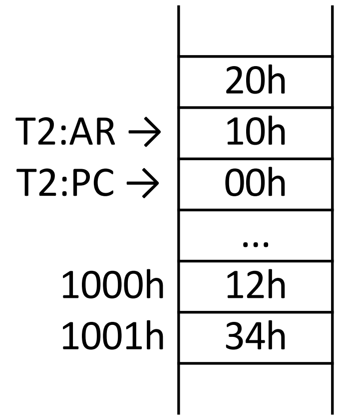
The opcode of the ADD instruction using direct addressing mode is 0 010 0000_2 = 20h. The ACC will be summed with the value at address 1000h and the result will be transferred back to the ACC.

As in other commands, the T0, T1 and T2 steps of this command are the same. The Execute cycle is as follows.

MICRO PROCESSING STEPS OF THE COMMAND	
T3.IDEC00.ADRMD2:	$TR_H \leftarrow M[AR]$, $AR \leftarrow AR+1$
T4.IDEC00.ADRMD2:	$TR_L \leftarrow M[AR]$, $PC \leftarrow PC+1$
T5.IDEC00.ADRMD2:	$AR \leftarrow TR$
T6.IDEC00.ADRMD2:	$DR_H \leftarrow M[AR]$, $AR \leftarrow AR+1$
T7.IDEC00.ADRMD2:	$DR_L \leftarrow M[AR]$
T8.IDEC00.ADRMD2:	$AC \leftarrow AC+DR$, $C \leftarrow C_{out}$,$SC \leftarrow 0$

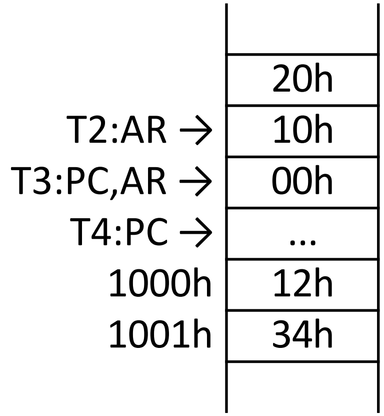
Steps T3 and T4 for Direct Mode

Values displayed by memory and registers at the end of the T2 cycle:



T3. IDEC00.ADRMD2: $TR_H \leftarrow M[AR]$, $AR \leftarrow AR+1$

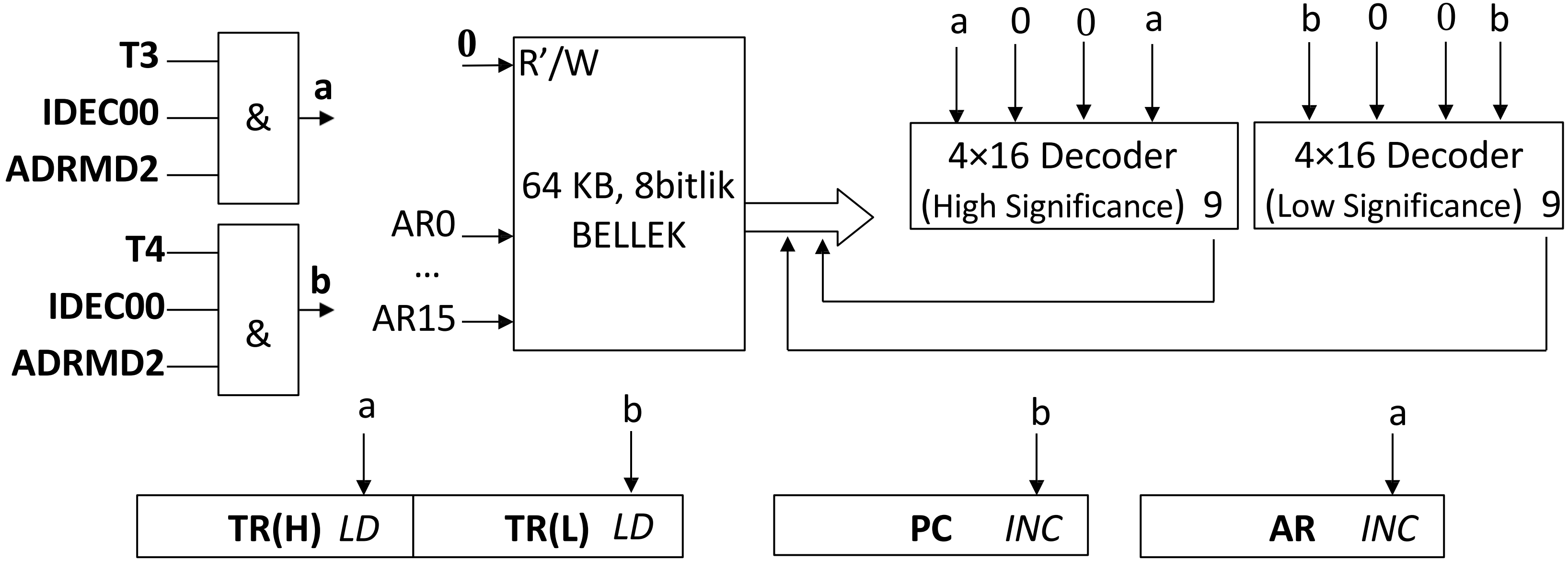
T4. IDEC00.ADRMD2: $TR_L \leftarrow M[AR]$, $PC \leftarrow PC+1$



Steps T3 and T4 for Direct Mode

T3. IDEC00.ADRMD2: $TR_H \leftarrow M[AR]$, $AR \leftarrow AR+1$

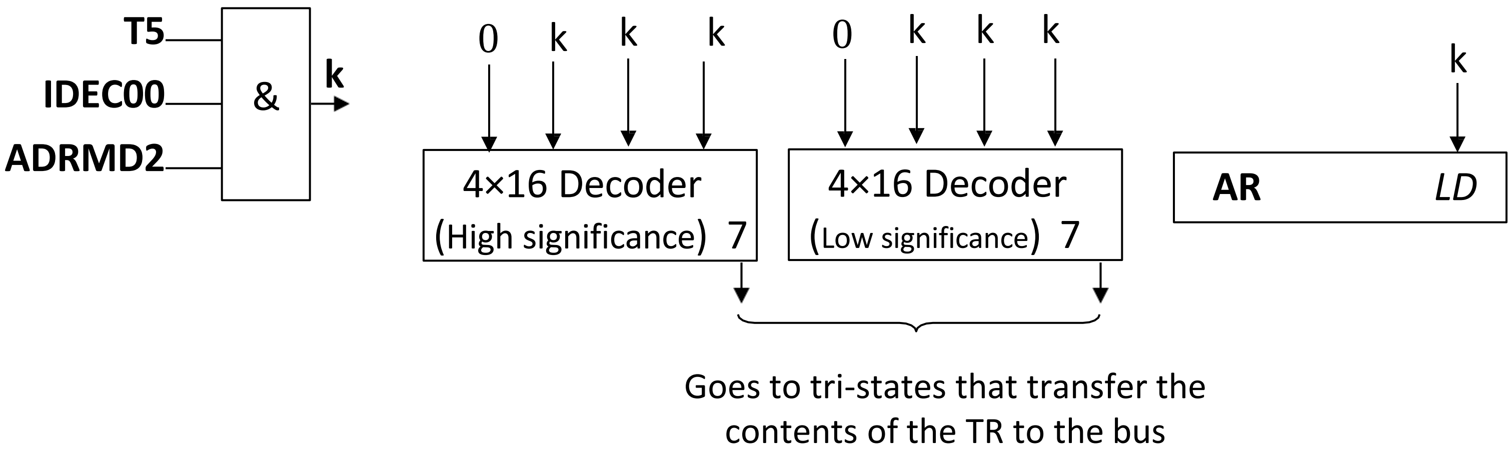
T4. IDEC00.ADRMD2: $TR_L \leftarrow M[AR]$, $PC \leftarrow PC+1$



T5 Step for Direct Mode

T5. IDEC00.ADRMD2: AR ← TR

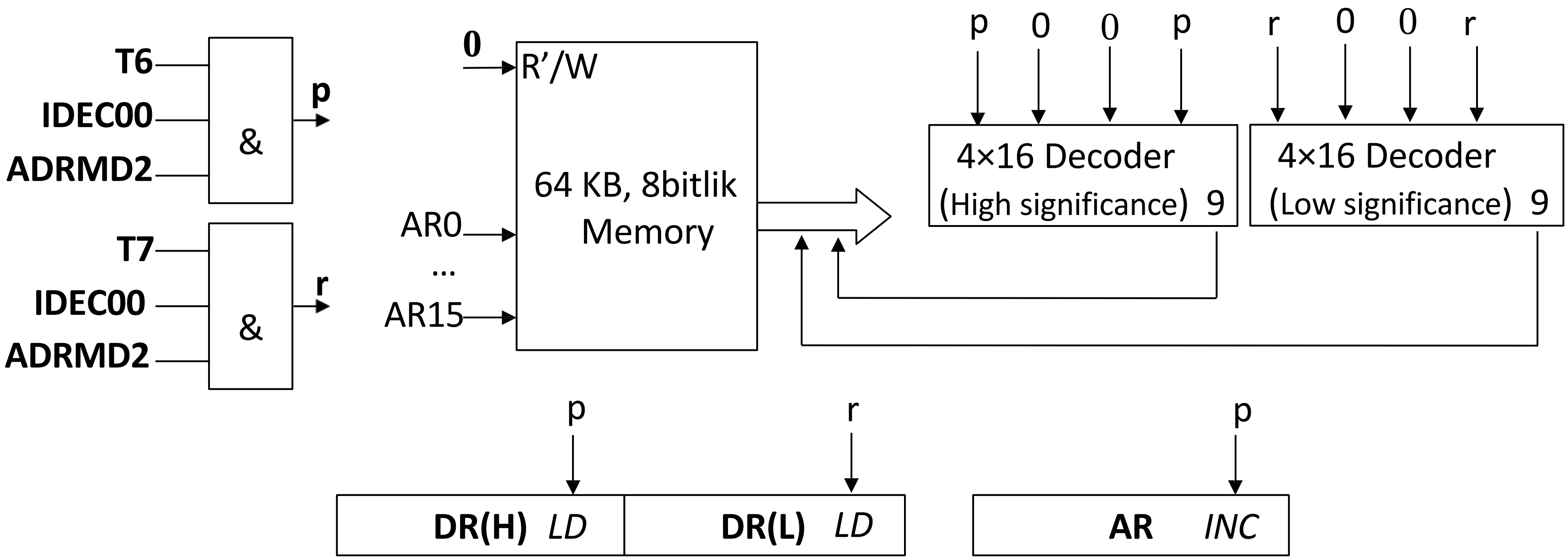
(Transfer was made to access the data in the location indicated by TR)



Steps T6 and T7 for Direct Mode

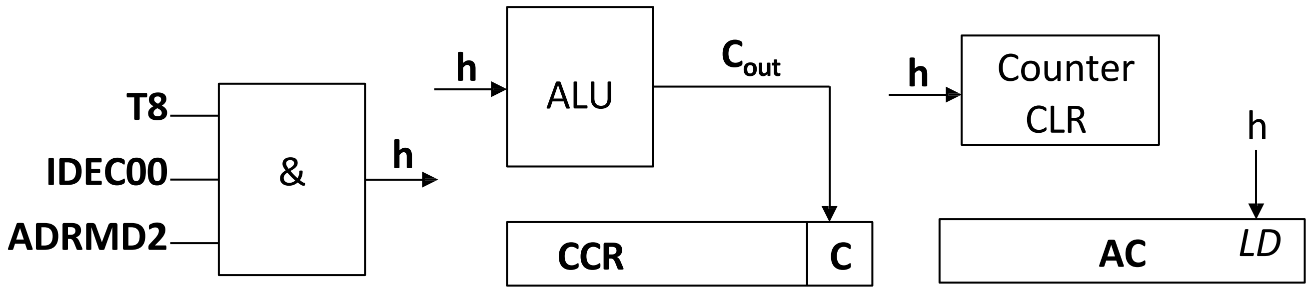
T6. IDEC00.ADRMD2 : $DR_H \leftarrow M[AR]$, $AR \leftarrow AR+1$

T7. IDEC00.ADRMD2 : $DR_L \leftarrow M[AR]$



T8 Step for Direct Mode

T8. IDEC00.ADRMD2: AC $\leftarrow$ AC+DR, C $\leftarrow$ C_{out} ,SC $\leftarrow$ 0



Example: Microoperation steps of ADD instruction with indirect addressing mode

Like the ADD (1000h) command. Here, parentheses are used in the address part, indicating indirect addressing mode. It is a 3 byte command.

The opcode of the ADD instruction using indirect addressing mode is 0 011 0000_2 = 30h. At address 1000h is the address of the operand. The ACC will be collected with the value at the specified address (0123h) and the result will be transferred back to the ACC.

	30h	
	10h	
	00h	
	...	
1000h	12h	Operand's address
1001h	34h	
	...	
1234h	01h	Operand's itself
1235h	23h	

MICRO PROCESSING STEPS OF THE COMMAND	
T3. IDEC00.ADRMD3:	$TR_H \leftarrow M[AR]$, $AR \leftarrow AR+1$
T4. IDEC00.ADRMD3:	$TR_L \leftarrow M[AR]$, $PC \leftarrow PC+1$
T5. IDEC00.ADRMD3:	$AR \leftarrow TR$
T6. IDEC00.ADRMD3:	$TR_H \leftarrow M[AR]$, $AR \leftarrow AR+1$
T7. IDEC00.ADRMD3:	$TR_L \leftarrow M[AR]$
T8. IDEC00.ADRMD3:	$AR \leftarrow TR$
T9.IDEC00.ADRMD3:	$DR_H \leftarrow M[AR]$, $AR \leftarrow AR+1$
T10. IDEC00.ADRMD3:	$DR_L \leftarrow M[AR]$
T11. IDEC00.ADRMD3:	$AC \leftarrow AC+DR$, $C \leftarrow C_{out}$,$SC \leftarrow 0$